

Can trees remember?

树有记忆吗？

2025 年 12 月 28 日 · 中英对照阅读

配套音频与原文: <https://www.bbc.co.uk/learningenglish/chinese/features/take-away-english/ep-251229>

人类能从过往经验中学习，从而延续有益的习惯、避免有害的行为。树也能通过世代交替不断进化，但它们也能像我们一样在自己的生命周期之内不断地适应周围的环境吗？随着气候变化加剧，许多地区的气候变化难以预测，这个问题的答案对自然环境的保护来说变得越来越重要。本文浅谈树是否能“记住”经验并从中学习。

词汇: trees 树

1 The English oak is one of Britain's most iconic trees. Broad and majestic , this tree supports more species than any other native tree in the UK, from blackbirds, to squirrels to butterflies. For centuries it has stood watching the landscapes change and our societies evolve. The question is, can a tree remember, learn and adapt to new environments within its lifetime? Does wisdom lie within its boughs ?

2 Trees provide a physical record of their life experiences in their growth rings . Dendrochronologists take samples from a tree trunk and are able to see whether a tree has been through a forest fire, periods of drought or a rainy season. But these markings are more than just battle scars . An organisation called Membra studies a number of UK tree species to see how they 'remember' and respond to environmental stress. Their research shows that environmental events do not change the tree's genetic code , but they can change how its genes react, which helps the tree adapt and defend itself against future drought, disease or climate extremes. So, trees don't have conscious memories of events like humans do - they remember through biological, structural changes.

3 That's a tree's defence in its own lifespan , but can their memories be passed down to future generations? Membra's research found that seedlings were more tolerant to drought when their parents had experienced repeated drought. And in one of the world's largest climate change experiments, published in Nature Climate Change by Richard Norby and colleagues, they pumped extra CO2 into an oak forest. Not only did the trees absorb more CO2, but the offspring of those trees also grew faster and were more resistant to disease, suggesting that tough conditions might make trees' offspring more resilient .

4 What does this all mean? There has been a wave of tree planting schemes throughout the UK, with aims to improve biodiversity and our defence against climate change. Fast-growing and short-lived tree species are often favoured in these schemes to quickly increase tree

numbers. But trees that live longer, like oaks, which have a greater memory and resilience, could be a better choice.

测验与练习

一、为本文选择总结句。

a. How can trees adapt to their environment? b. How trees are consciously aware of their surroundings. c. How trees reproduce.

二、根据文章内容选出正确的选项。

1. How does the author describe the English oak? a. in a positive way b. in a negative way c. in a neutral way
2. How do dendrochronologists learn about a tree's life? a. by checking the condition of its roots b. by observing its leaves throughout the year c. by taking a sample of its trunk and looking at its growth rings
3. Environmental events like repeated droughts can change the way the tree defends itself in future events. a. True b. False c. Not given
4. The trees that were exposed to extra CO₂ in the study published in Nature Climate Change were not able to absorb the extra CO₂. a. True b. False c. Not given
5. Based on the information we have about tree memory, which trees might be better for tree planting schemes? a. short-lived and fast-growing trees b. long-lived trees with greater memory c. short-lived and slow-growing trees

三、用下列单词完成对本文的总结。

A scientist that looks at a tree's 1) _____ to learn its age and its past experiences is called a 2) _____. The question is not about whether a tree is 3) _____, but whether it can learn from past events and become more 4) _____ against future environmental stresses. Beyond its own lifespan, research suggests that trees can pass down their learnings to future 5) _____.

growth rings resilient offspring dendrochronologist English oak biodiversity conscious

(答案见词汇表后)

词汇表

English oak

英国橡树 majestic 雄伟的, 壮观的 bough 大树枝 growth ring 年轮 dendrochronologist 树木年代学家 trunk 树干 battle scar 战斗留下的伤痕 genetic code 基因编码 conscious 有意识的 lifespan 生命周期 seedling 幼苗 offspring 后代 resistant 对...有抵抗能力的, 耐...的 resilient 适应力强的 biodiversity 生物多样性

答案

一、为本文选择总结句。

a. How can trees adapt to their environment?

二、根据文章内容选出正确的选项。

1. a . The English oak is one of Britain's most iconic trees. Broad and majestic , this tree supports more species than any other native tree in the UK , from blackbirds, to squirrels to butterflies.

2. c. Trees provide a physical record of their life experiences in their growth rings .
Dendrochronologists take samples from a tree trunk ...

3. a. True . Their research shows that environmental events do not change the tree's genetic code, but they can change how its genes react , which helps the tree adapt and defend itself against future drought , disease or climate extremes.

4. b. False. Not only did the trees absorb more CO2 ...

5. b. But trees that live longer , like oaks, which have a greater memory and resilience, could be a better choice .

三、用下列单词完成对本文的总结。

A scientist that looks at a tree's growth rings to learn its age and its past experiences is called a dendrochronologist . The question is not about whether a tree is conscious , but whether it can learn from past events and become more resilient against future environmental stresses. Beyond its own lifespan, research suggests that trees can pass down their learnings to future offspring .

内容版权归 BBC Learning English 所有，数据经 cooltv.top 获取，仅供个人学习使用。

原文：<https://www.bbc.co.uk/learningenglish/chinese/features/take-away-english/ep-251229>